Basic Electrical & Electronics Engineering Essentials

Course: Basic Electronics | Semester 2 | Contributor: Imlirenla Jamir

Kirchhoff laws, Thevenin Norton theorems, BJT biasing, and Operational Amplifiers.

Network Theorems Summary

* Thevenin theorem: open-circuit voltage Vth and looking-in resistance Rth.
* Maximum Power Transfer theorem: condition RL = Rth or RL = Zth conjugate.
* Superposition theorem applicability to linear bilateral networks.

Semiconductor Diodes & Op-Amps

* PN junction V-I characteristics, Zener voltage regulator breakdown mechanisms.
* Ideal Op-Amp properties: infinite input impedance, zero output impedance, virtual ground.
* Inverting and non-inverting amplifier closed-loop gain formulas.

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